

Crisis Communication Effectiveness on Social Media During COVID-19 Pandemic Rebound Periods: A Multi-Method Computational Analysis

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Abstract

This study examines which characteristics of COVID-19 vaccine-related tweets predicted audience engagement during the Delta variant rebound of July to September 2021. Drawing on a corpus of 71,845 tweets from a publicly available Kaggle dataset, the research applies a multi-method computational pipeline combining lexicon-based sentiment analysis (VADER), transformer-based sentiment analysis (Twitter-RoBERTa), neural topic modeling (BERTopic), and Zero-Inflated Negative Binomial regression with eleven predictors. Three core findings emerge. First, media richness, operationalized as the binary presence of an embedded URL, is the strongest content-level predictor of engagement, with an incidence rate ratio of approximately 4.6. Second, dialogic markers including question presence, hashtag count, and mention count are all negatively associated with engagement, contradicting the direct prediction of Dialogic Communication Theory. Third, sentiment has a small but consistent positive effect on engagement whose magnitude varies meaningfully between sentiment operationalizations. The findings support Media Richness Theory applied to a high-uncertainty crisis context, challenge the direct application of Dialogic Communication Theory to algorithmically mediated crisis communication, and establish methodological infrastructure for a planned doctoral research program on crisis communication effectiveness.

Keywords: Crisis Communication, COVID-19, Twitter, Sentiment Analysis, BERTopic, Negative Binomial Regression, Media Richness Theory

1 Introduction

The COVID-19 pandemic reshaped nearly every aspect of public life between 2020 and 2022, including the ways in which public health information circulates through society. For the first time in the history of global public health, hundreds of millions of people received real-time information about a life-threatening disease through the same communicative channels they used for personal conversation, entertainment, and commerce. Social media platforms such as Twitter, Facebook, and WhatsApp became simultaneously the primary conduit for authoritative health guidance, the primary venue for public anxiety, and the primary infrastructure for misinformation.

The present study investigates one specific dimension of this larger question: which characteristics of social media messages about COVID-19 vaccines predict audience engagement during the

Delta variant rebound of July to September 2021. The Delta rebound was a distinctive moment in the pandemic. Vaccines had been available for roughly six months in most Western countries, initial public enthusiasm had given way to vaccine fatigue, breakthrough infections were undermining the narrative that vaccination provided complete protection, and debates about booster doses and mRNA cardiac safety had moved to the center of public discourse.

The research is grounded in three communication theories: Media Richness Theory [1], Dialogic Communication Theory [2], and Situational Crisis Communication Theory [3]. Four hypotheses are tested: (H1) media richness increases engagement; (H2) dialogic markers increase engagement; (H3) positive sentiment increases engagement; (H4) findings are robust across sentiment operationalizations. As reported in the Results section, H1, H3, and H4 are supported, while H2 is rejected in a theoretically consequential way.

2 Related Work

Large-scale Twitter research on COVID-19 vaccine discourse has developed rapidly since early 2021. Lyu and colleagues [4] analyzed 1,499,421 tweets from March 2020 through January 2021, identifying 16 vaccine-related topics grouped into five themes and documenting upward trends in trust-related emotions. Yousefinaghani and colleagues [5] analyzed 4,552,652 tweets and found that positive-sentiment tweets received approximately 20 times more favorite hits than negative tweets. Xu and colleagues [6] extended this literature through December 2021 and identified distinct sentiment patterns for Chinese versus Western vaccines.

Two methodological gaps motivate the present study. First, the majority of existing work concludes before the Delta variant rebound of mid-2021, a period substantively distinct from the early pandemic phase. Second, existing studies typically rely on a single sentiment tool without robustness checks against transformer-based alternatives, which can produce meaningfully different sentiment distributions.

3 Data and Methods

3.1 Dataset

The data were obtained from the “All COVID-19 Vaccines Tweets” dataset maintained by Gabriel Preda on Kaggle [9], distributed under the CC0 public domain license. The raw corpus spans December 2020 to November 2021. The dataset was restricted to the window between July 1 and September 30, 2021 (the Delta rebound). After removing exact duplicates and records with missing text, the final working corpus contains **71,845 unique tweets**. Two geographic subsets were identified: 3,838 tweets from the United States and 1,606 tweets from the United Kingdom.

3.2 Sentiment Analysis

Two complementary sentiment models were applied. The primary model is VADER [7], a rule-based sentiment analyzer designed for social media text that produces a compound score bounded between -1 and $+1$. The secondary model is the `cardiffnlp/twitter-roberta-base-sentiment-latest` checkpoint [8], a RoBERTa-base transformer pre-trained on 124 million tweets and fine-tuned for three-way sentiment classification. The cross-model Pearson correlation is $r = 0.456$ ($p < 0.001$), indicating moderate agreement consistent with the operational differences between lexicon-based and learned sentiment representations.

3.3 Topic Modeling

Topic modeling was performed using BERTopic [10], which combines sentence embeddings, dimensionality reduction, density-based clustering, and a class-based TF-IDF topic representation. Embeddings were computed with `all-MiniLM-L6-v2`, reduced to five dimensions via UMAP, and clustered with HDBSCAN (`min_cluster_size = 150`). The initial fit produced 105 topics with 20,730 outliers; after outlier reduction and hierarchical topic merging, the final structure contains 19 substantive topics plus 3,531 residual outliers.

3.4 Feature Engineering

Five feature families were constructed: (1) theoretical variables (sentiment, media richness via URL presence, dialogic markers including hashtags, mentions, and question presence); (2) medical named-entity features via a dictionary-based procedure covering 17 surface forms mapped to 8 vaccine brands and 25 side-effect terms; (3) temporal features (hour, day of week, weekend indicator); (4) stylistic intensity (emoji count, exclamation count, URL count); and (5) the outcome variable, defined as the sum of favorites and retweets.

3.5 Statistical Modeling

Three regression specifications were fit. The baseline Negative Binomial model uses five theoretical predictors with VADER sentiment. An upgraded specification adds RoBERTa sentiment and tweet length. The final specification is a Zero-Inflated Negative Binomial (ZINB) model with eleven predictors in the count component and tweet length in the inflation component. Estimation used the BFGS optimizer with a warm-start strategy initialized from the fitted NB coefficients.

4 Results

4.1 Model Fit

Table 1 reports model comparison statistics. The final ZINB specification improves AIC by 4,348 points relative to the baseline, providing strong statistical support for the upgraded specification.

Model	Predictors	AIC
Baseline NB (VADER)	5	259,082.23
Upgraded NB	7	258,520.34
Final ZINB (RoBERTa)	11	254,733.93

Table 1: Model comparison across three specifications.

4.2 Regression Coefficients

Table 2 reports the count-component coefficients from the final ZINB specification.

Predictor	Coef.	IRR	<i>p</i>
Media richness (URL)	+1.517	4.56	< .001
RoBERTa sentiment	+0.668	1.91	< .001
Side-effect mention	+0.184	1.20	< .05
Vaccine count	+0.093	1.10	< .001
Weekend indicator	+0.064	1.07	< .05
Mention count	−0.116	0.89	< .001
Hashtag count	−0.139	0.87	< .001
Question presence	−0.271	0.76	< .001

Table 2: Final ZINB count-component coefficients ($n = 71,845$).

4.3 Topic Structure

The BERTopic pipeline produced 19 substantive topics. The four largest topic clusters are Indian vaccination booking notifications (14,874 tweets), Moderna-related content (11,943 tweets), Covaxin and OCGN stock discussion (7,205 tweets), and Sinopharm/Sinovac discussion (5,093 tweets). Medical-safety topics including cardiac side effects (*myocarditis*) are smaller in volume but attract disproportionate engagement per tweet.

4.4 Robustness

A direct comparison of the Negative Binomial regression under VADER versus RoBERTa sentiment with identical non-sentiment predictors preserves the direction and approximate magnitude of every theoretical effect. Media richness has an IRR of 4.71 under VADER and 4.58 under RoBERTa. Question, hashtag, and mention effects remain negative and significant under both specifications. Sentiment magnitude shifts (IRR = 1.37 under VADER, 1.91 under RoBERTa), but direction is preserved.

5 Discussion

The empirical findings converge on a single overarching pattern: in the context of a pandemic rebound, richer communication outperforms leaner communication by a large margin, while communication that signals dialogue underperforms on every measurable dimension.

The 4.6-fold multiplier on media richness is substantively large and robust across specifications. This finding supports Media Richness Theory [1] in a crisis context: under uncertainty, audiences reward messages that carry an informational promise (a linked article, dataset, or primary source) over unadorned text.

The negative effect of dialogic markers is theoretically consequential. Under Dialogic Communication Theory [2], tweets carrying questions, hashtags, or mentions should outperform broadcast-style content. The opposite is observed. Three interpretations are possible: (1) under crisis conditions, audiences prioritize information over conversation; (2) dialogic markers may correlate with account-level characteristics that the present data cannot control for (follower counts are unavailable); (3) algorithmic mediation may have altered the role of dialogic markers since the original 1998 formulation of the theory.

6 Conclusion

This research contributes to crisis communication scholarship on three dimensions. First, it adds empirical coverage of the under-researched Delta rebound window. Second, it demonstrates the feasibility of a rigorous multi-method computational pipeline within the constraints of a master’s thesis research project. Third, it challenges the direct application of Dialogic Communication Theory to algorithmically mediated crisis communication. The methodological infrastructure developed here is portable to other crisis contexts and is intended to form the foundation of a follow-up doctoral program at ITMO University.

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